Junghyun Choi[1]
Jaebum Choo[1]
Hoeil Chung[2]
Dae-Gab Gweon[3]
Jeunghee Park[4]
Hyo Jin Kim[5]
Sangyong Park[6]
Chil-Hwan Oh[6]

Direct Observation of Spectral Differences Between Normal and Basal Cell Carcinoma (BCC) Tissues Using Confocal Raman Microscopy

[1] *Department of Applied Chemistry, Hanyang University, Ansan 426-791, South Korea*

[2] *Department of Chemistry, Hanyang University, Seoul 133-791, South Korea*

[3] *Department of Mechanical Engineering, KAIST, Daejeon 305-701, South Korea*

[4] *Department of Chemistry, Korea University, Jochiwon 339-700, South Korea*

[5] *Department of Pharmacy, Dongduk Women's University, Seoul 136-174, South Korea*

[6] *Department of Dermatology, Korea University, Seoul 152-703, South Korea*

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Abstract: *Raman spectroscopy has strong potential for providing noninvasive dermatological diagnosis of skin cancer. In this study, confocal Raman microscopy was applied to the dermatological diagnosis for one of the most common skin cancers, basal cell carcinoma (BCC). BCC tissues were obtained from 10 BCC patients using a routine biopsy and used for confocal Raman measurements. Autofluorescence signals from tissues, which interfere with the Raman signals, were*

Correspondence to: Jaebum Choo (jbchoo@hanyang.ac.kr) or Chil-Hwan Oh (choh@korea.ac.kr)

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greatly reduced using a confocal slit adjustment. Distinct Raman band differences between normal and BCC tissues for the amide I mode and the PO_2^- symmetric stretching mode showed that this technique has strong potential for use as a dermatological diagnostic tool without the need for statistical treatment of spectral data. It was also possible to precisely differentiate BCC tissue from surrounding noncancerous tissue using the confocal Raman depth profiling technique. We propose that confocal Raman microscopy provides a novel method for dermatological diagnosis since direct observations of spectral differences between normal and BCC tissues are possible. © 2005 Wiley Periodicals, Inc. Biopolymers 77: 264–272, 2005

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Keywords: *confocal Raman microscopy; skin cancer; basal cell carcinoma; noninvasive; diagnosis*

INTRODUCTION

Advances in spectroscopic technology over the past few decades have had a significant impact on the area of cancer research. Technologies such as Fourier transform infrared (FTIR), Raman, and confocal fluorescence microscopy have been used to elucidate the origin and progression of cancer.[1–6] These technologies have also been used for the early detection of cancer and biochemical changes in cells and tissues causing cancer.[7,8] Among these spectroscopic techniques, Raman spectroscopy has attracted considerable attention for medical diagnosis since it is nondestructive, does not require a sample preparation process, and provides detailed information about the molecular structure of normal and diseased tissues.

Recently, the incidence of skin cancer has dramatically increased due to the excessive exposure of skin to UV radiation caused by ozone layer depletion, environmental contamination, and so on. If detected early, skin cancer has a cure rate of 100%. Unfortunately, early detection is difficult because diagnosis is still based on morphological inspection by a pathologist. There are two common skin cancers: basal cell carcinoma (BCC) and squamous cell carcinoma (SCC). Both BCC and SCC are nonmelanoma skin cancers, and BCC is the most common skin neoplasm.[9–12] The accurate detection of BCC has attracted much attention from clinical dermatologists since it is difficult to distinguish BCC tissue from surrounding noncancerous tissue.[13–16] The routine diagnostic technique used for the detection of BCC is pathological examination of biopsy samples. This involves removal of tissue from suspected areas, which is then sliced and stained to enable the pathologist to identify morphological abnormalities. This method relies upon a subjective judgment, which is dependent on the level of experience of the individual pathologist and can lead to the excessive biopsy of tissues.[2] Thus, a fast and accurate diagnostic technique for the initial screening and selection of lesions for further biopsy is needed.

Raman spectroscopy has the potential to resolve this problem.[17] It can be applied to provide a more accurate medical diagnosis to distinguish BCC tissue from surrounding normal tissue. Recently, several research groups carried out BCC detection using Raman spectroscopy. Gniadecka et al.,[18–20] Nunes et al.,[21] Schrader et al.,[22] Fendel et al.,[23] Skrebova et al.,[24,25] and Barry et al.[26] used Fourier transform (FT)–Raman spectroscopy to distinguish between BCC and surrounding normal tissues. Nijssen et al.[10] demonstrated pseudo-color Raman maps based on FT–Raman spectra obtained from a frozen tissue section of BCC. In this work, they showed that the pseudo-color Raman map is closely related to the microscopic image of the hematoxylin and eosin (H&E)-stained sample. In previous studies using FT–Raman spectroscopy, a long wavelength excitation laser (1064 nm Nd:YAG laser or 850-nm titanium–sapphire laser) has been applied to minimize autofluorescence from skin tissue. However, the longer wavelength laser has poor Raman scattering intensities compared to the shorter excitation wavelength laser. As a result, previously reported Raman spectra on BCC tissues show poor signal-to-noise ratios and require statistical treatment of spectroscopic data, such as principal component analysis (PCA),[21] K-mean clustering analysis (KCA),[10] and neural networks,[18–20] to differentiate between Raman signals of cancerous and normal tissues. For the practical application of cancer diagnosis, however, direct observation of spectral differences without any statistical treatment of spectral data is essentially required.

In the present study, the confocal Raman technique, using a shorter wavelength argon ion laser at

514.5 nm, was applied to the dermatological diagnosis of BCC. To effectively remove autofluorescence, which strongly interferes with Raman signals, the confocal technique was used. Confocal Raman microscopy has been known to exhibit a superior rejection of fluorescence.[27] This arises from the fact that the laser is focused on such a small point that the laser flux in the sampling volume is high enough to quench fluorescence in a fraction of time often needed in conventional Raman spectroscopy. Second, the excited electron, along with its slit, can migrate from one molecule to the other so that a molecule outside the focal volume may emit the fluorescence photon. These photons are blocked by the confocal slit. Consistent spectral differences between normal and cancerous tissues were observed for 10 BCC samples investigated using confocal Raman microscopy. Furthermore, confocal Raman depth profiling was performed to differentiate BCC tissue from surrounding noncancerous tissue; it is important to determine the biopsy area in a surgical operation. Here, we report the potential for confocal Raman microscopy to be used as a direct diagnostic tool for precancerous and noncancerous lesions.

MATERIALS AND METHODS

Skin Tissue Preparations

Skin tissue samples were obtained from the Dermatologic Department of Korea University Hospital in Korea. BCC tissues used for confocal Raman measurements were obtained from 10 patients using a routine biopsy. Cross sections 20-μm thick were cut with a microtome at $-20°C$, and frozen sections were stored in liquid nitrogen before use. Two thin sections were cut for the experiments. One section was used for confocal Raman profiling experiments and the other section was stained with H&E. Then the stained section was sent to an expert pathologist for a routine cancer diagnosis. The H&E section was also used as a Raman reference to locate the boundaries between the different skin layers in the unstained section.

Confocal Raman Measurements

Confocal Raman measurements were performed using a Renishaw 2000 Raman microscope system. This system is set up to ensure that light from the region of the sample that is in best focus to the Raman system is accepted, while light from the adjacent regions, which will be out of focus, is rejected. In this way, spectral interference or contamination from the surrounding regions is greatly minimized, compared to nonconfocal operation. A Spectra Physics argon ion laser operating at $\lambda = 514.5$ nm was used as the excitation source with a laser power of approximately 20 mW. The Rayleigh line was removed from the collected Raman scattering by a holographic notch filter located in the collection path. Raman scattering was detected using a charge-coupled device (CCD) camera at a spectral resolution of 4 cm^{-1}. An additional CCD camera was fitted to an optical microscope to obtain visual images of skin tissue samples. A two-slit confocal arrangement was used to reduce background Raman scattering from the unfocused laser beams. All Raman spectra were measured in the confocal mode to remove out-of-focus signals.[28,29] In the Raman system, the function of the pinhole was replaced by the cooperation of the entrance slit and the pixels in the CCD detector. The first confocal slit was set to a width of 15 μm. Usually the slit is set to between 50 and 100 μm in nonconfocal usuage. The image area of CCD used for capturing grating spectra was also reduced to 4 pixels in the confocal mode. The signal was then collected from only two pixel rows on the

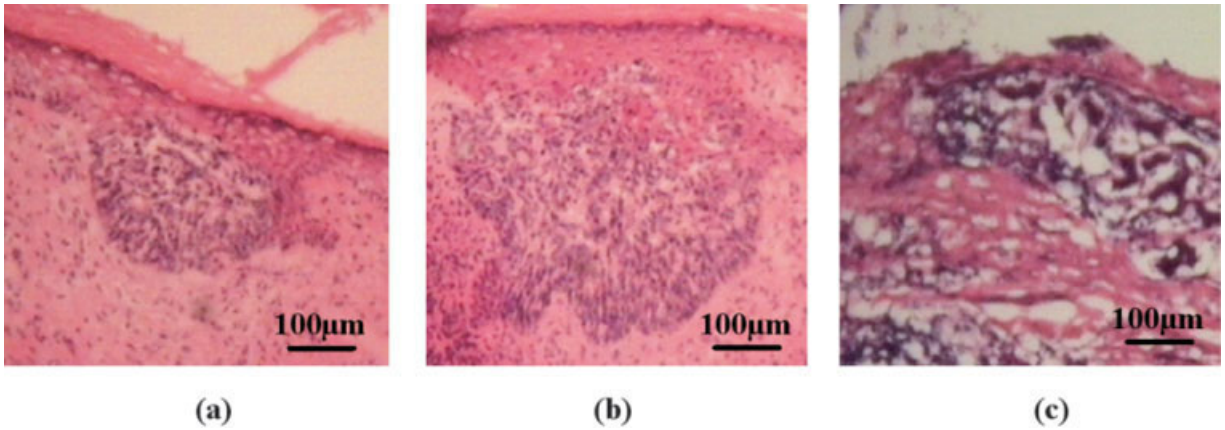

FIGURE 1 Microscopic images of H&E-stained sections for skin tissue collected from three different patients (a, b, and c). Dark blue regions are malignant tissues (BCC) and lighter eosinophilic regions are normal tissues.

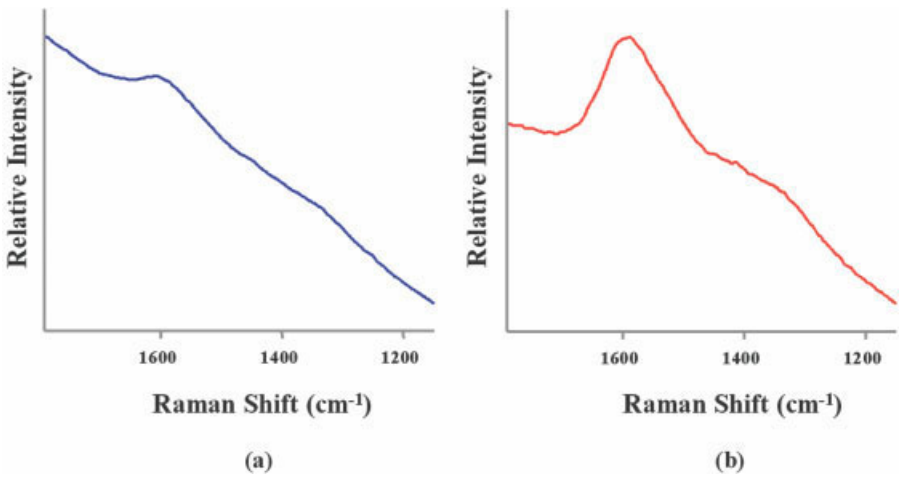

FIGURE 2 Raman spectra of BCC tissue: (a) nonconfocal mode Raman spectrum; (b) confocal mode Raman spectrum.

CCD detector, creating a virtual second slit that was aligned perpendicular to the spectrometer slit. In this way, stray background light due to any out-of-focus regions of the skin tissue sample was effectively removed.

RESULTS AND DISCUSSION

Figure 1 shows microscopic images of the H&E-stained tissues obtained from three different skin cancer patients. Dark regions are BCC tissues and lighter regions are normal tissues. This pathological examination of biopsy samples provides a standard for the Raman spectroscopic identification of BCC. To confirm the necessity of the confocal technique in Raman measurements, the Raman spectra of BCC were measured using two different modes: a nonconfocal mode and a confocal mode. Figure 2 shows a comparison of the Raman spectra of BCC tissues measured using these two different modes: the nonconfocal mode in Figure 2a and the confocal mode in Figure 2b. Fluorescence interference was very high in the nonconfocal mode Raman spectrum, whereas interference signals from BCC tissues were greatly reduced in the confocal mode spectrum when a 15-μm wide confocal slit was used to remove any unfocused scattering signals. Therefore, the confocal technique is very effective at reducing autofluorescence signals from tissues.

Figure 3 shows the confocal Raman spectra for the skin sample shown in Figure 1a. A clear distinction was found between normal and BCC tissues by confocal Raman microscopy. Many differences in these spectra arise from the intensity changes in the 1000–1700-cm^{-1} region. In previous studies,[10,18–20,21] trivial spectral differences between normal and BCC tissues were found only in the amide III mode at the 1250–1350 cm^{-1} region and in the PO_2^- stretching mode at the 1000–1100-cm^{-1} region. In our confocal Raman spectra, however, clear spectral differences were found in four characteristic Raman bands in this region. As shown in Figure 3, it is possible to make a visual diagnosis from the spectra without any statistical treatment. For a more detailed comparison between the two Raman spectra, the Raman bands of BCC tissue were decomposed using Gaussian curve-fitting analysis (dot line). The broad overlap of bands in the 1200–1800-cm^{-1} region was decomposed into three characteristic bands. Initial conditions of the band center and width for the curve fitting of BCC

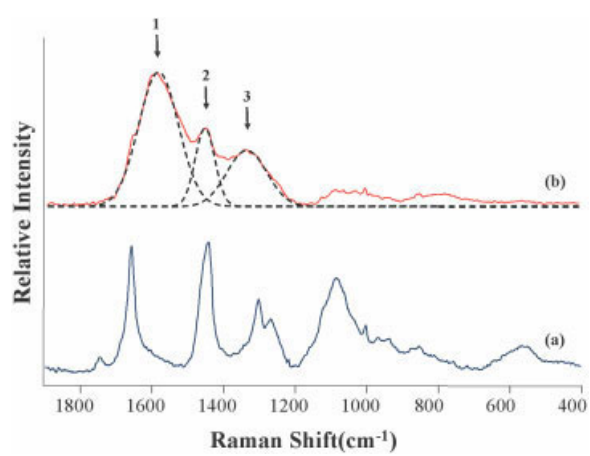

FIGURE 3 Confocal Raman spectra of skin tissue in the 1000–1700-cm^{-1} region: (a) Raman spectrum of normal tissue; (b) curve-fitting Raman spectrum of BCC tissue.

Table I Initial Conditions of Band Center and Width for the Curve Fitting of BCC Tissue

Initial Condition	Amide I Mode (1)	Protein and Lipid Mode (2)	Amide III Mode (3)
Band center	1583	1454	1337
Band width	124	69	120

tissue are listed in Table I. The assignments of characteristic Raman bands for the normal and BCC tissues are listed in Table II. As shown in Figure 3 and Table II, the amide I band at 1656 cm^{-1} was shifted to the lower frequency and its bandwidth was greatly broadened for BCC tissue, whereas the amide III band at 1302 cm^{-1} was shifted to a higher frequency and its intensity decreased. These results are similar to previous Raman data reported by Gniadecka et al.,[18–20] but our confocal spectra show much more distinct differences between normal and BCC tissues. The Raman band changes in amide I and amide III modes are closely related to the structural changes in secondary protein between normal and BCC tissues. The CH_2 deformation mode at 1441 cm^{-1} for lipids and proteins shifted to a higher frequency and its intensity decreased. The PO_2^- symmetric stretching mode at 1085 cm^{-1} for phospholipids and nucleic acids shifted to a lower frequency and its intensity was greatly reduced.

Consistent spectral differences between normal and BCC tissues were observed for 10 tissue samples. Principal component analysis (PCA) was performed to clearly discriminate the Raman spectra for normal and BCC tissues. Before performing PCA, the spectral baseline for the 400–1700-cm^{-1} region was corrected and zeroed, and all the spectral data in this range were used for the PCA. The results are displayed in Figure 4. The discrimination between normal and BCC tissues are very clear and a discrete boundary can be seen in this figure. The F test, based on the one-way multivariate analysis of variance, was performed to estimate the value of statistical discrimination. The F value was calculated to be 329.5, and this means that the two groups are significantly different from each other at the confidence level of 95%.

Table II Observed Confocal Raman Frequencies of Normal and BCC Tissues

Vibrational Descriptions	Normal Tissue (cm^{-1})	BCC Tissue (cm^{-1})[a]
Amide I mode	1656	1589
Lipid and protein mode (CH_2 deformation)	1441	1450
Amide III mode	1302	1328
Phospholipid and nucleic acid mode (PO_2^- symmetric stretch)	1085	1048

[a] Assigned frequencies based on curve-fitting analysis.

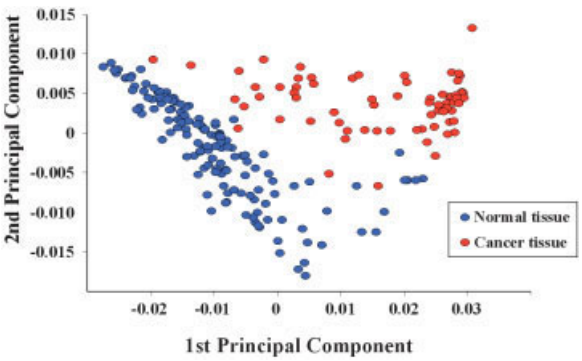

FIGURE 4 Plot of PCA for 10 tissue samples in the 1000–1700-cm^{-1} range.

Figure 5 shows the Raman spectra for normal and BCC skin tissues from the three different patients in Figure 1. The Raman spectra for the three different normal tissues are very similar to each other and to those previously reported. The Raman spectra of BCC tissues from the three patients also show very good reproducibility. Direct spectral differences between normal and BCC tissues can also be seen (Figure 5). In particular, the enhancement of the amide III band and the disappearance of the PO_2^- stretching band in BCC tissue can be used for the direct dermatological diagnosis of a skin cancer.

Another important factor in dermatological diagnosis is how precisely BCC can be distinguished from surrounding noncancerous tissue using Raman spectroscopy.[30,31] Raman spectroscopy has the potential to be used as a fast and accurate diagnostic tool for the screening and selection of a biopsy area. In order to confirm this, we performed confocal Raman depth profiling for BCC and surrounding tissue. Figure 6a shows the enlarged microscopic image of H&E-stained tissue in Figure 1a. A skin biopsy was performed to the perpendicular direction of the skin surface, and the spectral measurements were acquired from the epidermis to the dermis direction (z-axis in Figure 6a). The thick horizontal bars indicate the Raman sampling spots. Raman spectra of BCC tissue were measured at ten different spots with intervals of 30–40 μm from the epidermis side to the dermis side. The profiling distance was accurately adjusted by a

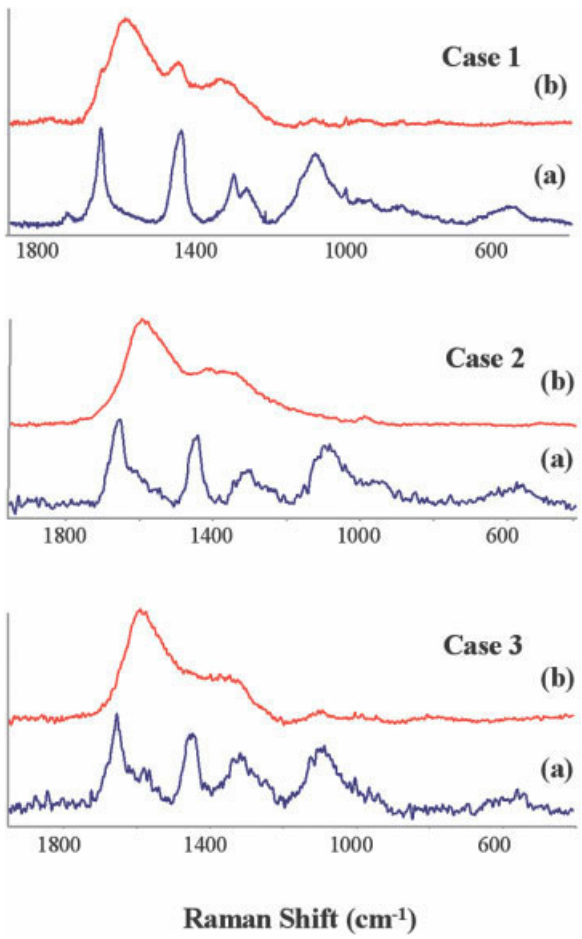

FIGURE 5 Confocal Raman spectra of normal and malignant tissue from three different patients: (a) normal tissue; (b) BCC tissue.

high-precision computer controlled stage. Figure 6b shows a set of Raman spectra of skin tissues obtained from the same specimen but at different spots intersecting the border between the cancerous and normal regions. For example, the spectrum "G" shows a typical pattern of cancerous tissue and the spectrum "J" shows that of a normal tissue. Raman spectra were measured for the frozen tissues since the fluorescence from the H&E-stained tissue strongly interferes with the Raman signals. The H&E-stained image was just used as a reference since we could not see the difference between normal and BCC areas with the image of frozen tissues for the Raman measurements. After the Raman measurements, some of the samples were stained again in order to confirm the positions of Raman measurement spots.

As shown in Figure 6, the strong PO_2^- symmetric stretching band centered at 1085 cm^{-1} was observed at the normal epidermis (A and B) and dermis (I and J) areas. On the other hand, the strong amide I band centered at 1589 cm^{-1} was observed in the BCC (D, E, F, and G) area. On the border between normal and BCC tissues (C and H), intermediate intensities of the PO_2^- band were observed. In order to confirm the reproducibility of the confocal Raman technique for dermatological diagnosis, we performed confocal Raman profiling experiments for the other skin tissues in Figure 1. Figure 7 shows the Raman profiling data for the second skin tissue sample in Figure 1b. In this case, very similar patterns of Raman bands to the first case were observed. In the epidermal area (A, B, and C), the strong BCC amide I band was not observed and the characteristic strong PO_2^- band of the normal tissue was found. For the BCC area (D, E, F, G, H, and I), however, strong amide I bands were observed, but strong PO_2^- bands for normal tissue were also found in some cases (D, F, G and H). When the H&E-stained image was compared with the corresponding Raman spectra, it appeared that the BCC had not homogeneously spread out. The last spectrum (J) in the dermal area shows the characteristic Raman bands of noncancerous tissue. Figure 8 displays spectra for the sample in Figure 1c. At the dermal area (A) and normal dermis area (H, I, J, and K), typical Raman spectra of normal tissue were seen, and at the BCC dermis area (D, E, F, and G) characteristic Raman bands for BCC tissues were observed. In this sample, however, it is interesting to note that mixed Raman bands were observed on the border (B and C) between normal and BCC tissues. The spectral changes observed from A to D show that the amide I band for the normal tissue has shifted to the band for the BCC tissue. Clear intensity changes in the PO_2^- and amide III bands from the normal side to the cancerous side of the tissue can also be seen. These results show that by applying the confocal Raman depth profiling technique to human skin tissue, it is possible to precisely determine cancerous tissue from surrounding noncancerous tissue.

CONCLUSIONS

Raman spectroscopy has been widely used to differentiate BCC tissue from healthy surrounding tissue. In earlier studies, the FT–Raman spectroscopic technique, using a long-wavelength near-infrared laser, has been applied to minimize autofluorescence from human tissues. In these cases, however, the Raman intensity is relatively weak due to a poor scattering efficiency, even though the spectrum has accumulated for a long time. As a result, statistical treatment of collected Raman data, using PCA analysis or neural network, is required to differentiate BCC tissue from

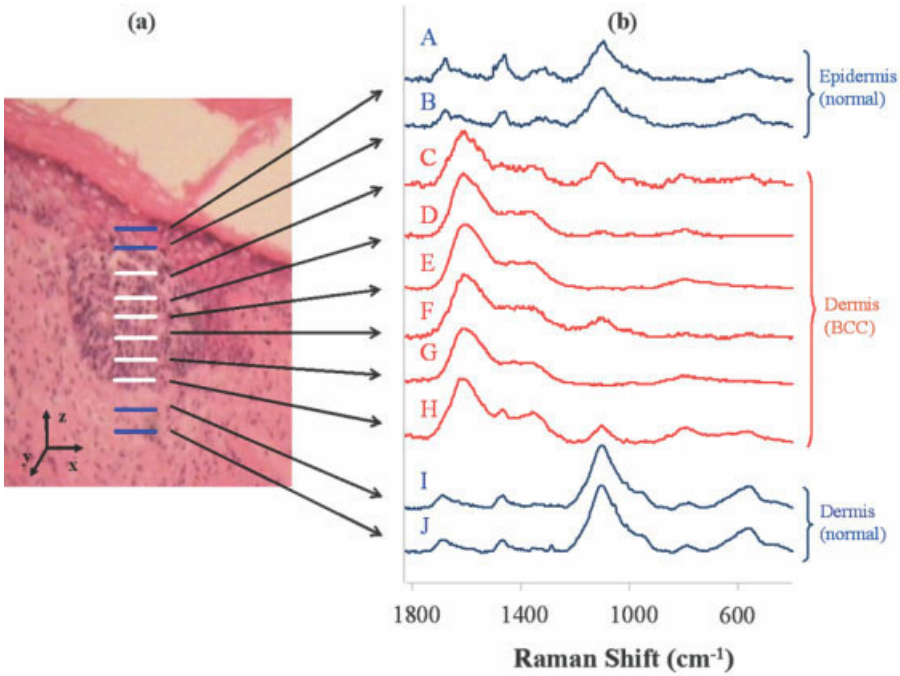

FIGURE 6 Confocal Raman profile of skin tissue [Figure 1(a)] for the epidermis and dermis areas with the intervals of 30–40 μm: (a) enlarged microscopic image of H&E-stained skin tissue; (b) corresponding Raman spectra at ten different spots indicated in (a).

normal tissue. In the present work, we found that confocal Raman microscopy, using the 514.5-nm argon ion laser, can be successfully applied to the dermatological diagnosis of BCC skin cancers. The out-of-plane autofluorescence signals from tissues could be efficiently removed by the confocal tech-

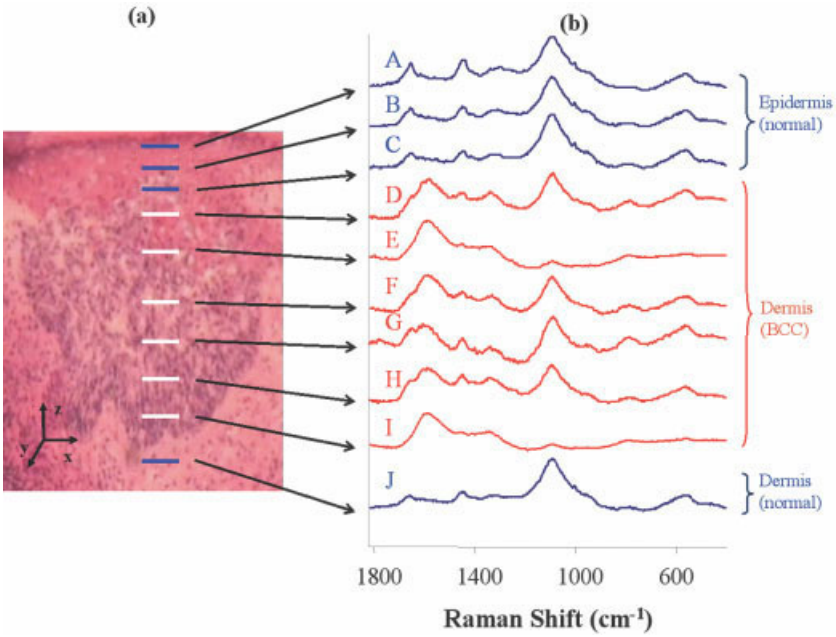

FIGURE 7 Confocal Raman profile of skin tissue [Figure 1(b)] for the epidermis and dermis areas with the intervals of 30–40 μm: (a) enlarged microscopic image of H&E-stained skin tissue; (b) corresponding Raman spectra at ten different spots indicated in (a).

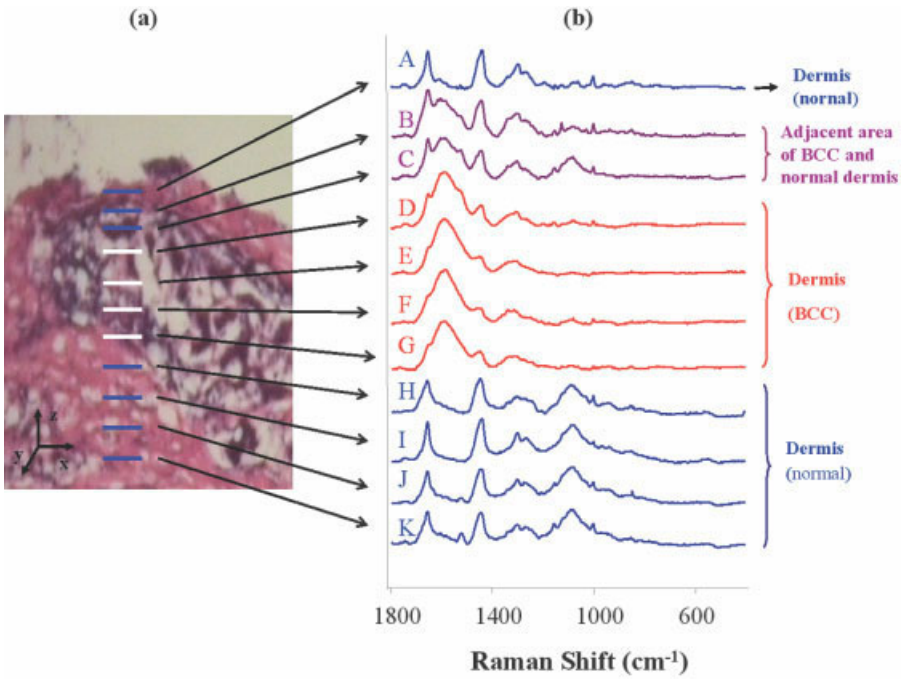

FIGURE 8 Confocal Raman profile of skin tissue [Figure 1(c)] for the dermis area with the intervals of 30–40 μm: (a) enlarged microscopic image of H&E-stained skin tissue; (b) corresponding Raman spectra at eleven spots indicated in (a).

nique. In particular, distinct Raman band differences between normal and BCC tissues for the amide I mode and the PO_2^- symmetric stretching mode show that this confocal Raman technique has strong potential as a dermatological diagnostic tool without the need for any statistical treatment of data.

A fast and accurate BCC diagnostic tool for the screening and selection of biopsy is important in Moh's micrographic surgery. Therefore, confocal Raman depth profiling could be used to accurately determine the BCC area from healthy surrounding tissue, making it possible to precisely differentiate a cancerous area from surrounding noncancerous tissues. Mixed amide I bands of normal and BCC tissues around the borderline, as well as irregular patterns of PO_2^- bands of normal tissue caused by inhomogeneous extension of BCC in the dermis area, can also be detected. We predict that this confocal Raman profiling technique has strong feasibility as a dermatological diagnostic tool.

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